

Linking Rationale to Decision on Internet Standards

A Retrieval-Based Approach Using Synthetic Data

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Some key concepts are as follows:

RFCs (an acronym for their historical title, "Requests for Comments") are the formal documents that define Internet standards.

Decision refers to the normative and descriptive statements within an RFC that specify the chosen mechanisms, structures, and operational behaviors of a protocol.

Rationale is a set of explanatory statements that account for why specific textual elements in an RFC appear in their current form.

In short, **Rationale** explains, supports, and justifies the **Decision**.

A Toy Example: Rationale → Decision

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Rationale

To simplify implementation across platforms, we avoid using platform-specific cryptographic primitives.

⇒

Decision

The protocol **SHOULD** use standardized AES encryption with a fixed key length.

- Rationale explains *why* a choice is made.
- Decision specifies *what* the standard requires.

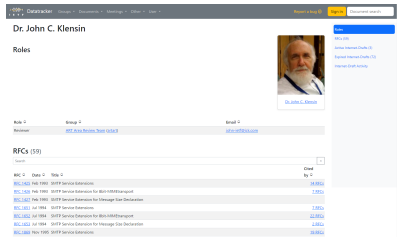
Decisions are scattered across roughly 10 000 RFCs, each with multiple draft versions, and individual documents often span dozens of pages.

Rationales are buried in nearly 40 years of IETF mailing-list archives, dating back to the mid-1980s.

However, there is no explicit linkage between the two sources.

For a single RFC decision, manually tracing historical email threads is extremely time-consuming, error-prone, and practically infeasible at scale.

- Data are abundant, but labeled rationale–decision links remain scarce.
- Large-scale annotation is expensive and requires substantial domain expertise.
- This expertise is limited to a small number of specialists.



To address these challenges, we propose a two-step framework that combines synthetic decision reconstruction and context-aware information retrieval.

- We leverage LLMs to reconstruct the decision implied in an email discussion, generating a text snippet that corresponds to the relevant section of the associated RFC.
- We design a *rationale-to-decision* retrieval, where the input is an email discussion and the target is the corresponding decision.

I. Synthetic Data: Generation Direction

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Rationale

To simplify implementation across platforms, we avoid using platform-specific cryptographic primitives.

⇒

Decision

The protocol **SHOULD** use standardized AES encryption with a fixed key length.

Key Observation

Forward generation (Rationale ⇒ Decision) is easier and more reliable than reverse inference (Decision ⇒ Rationale).

I. Synthetic Data: Prompt Strategy

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Try-Then-Infer

Generate the synthetic decision directly from the email content.

Judge-Then-Infer

- First, judge whether the email contains substantive technical discussion.
- If yes: preserve quoted specification text when present; otherwise synthesize a plausible statement.
- If no: return an empty string.

Anchor-Then-Infer

Use subject-line anchors (e.g., Re: [Technical Errata Reported] RFC8209 (8653)) to identify the relevant RFC context before generation.

I. Synthetic Data: Prompt Strategy Selection

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Try-Then-Infer

Serves as a baseline.

Judge-Then-Infer

Yields more consistent and accurate results.

Anchor-Then-Infer

Often failed to produce satisfactory outputs, contrary to expectations, likely due to the high degree of interconnectedness among RFCs.

I. Synthetic Data: LLM Variants

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LLMs: Qwen-32B and Llama-70B

Verbose email (with quotations)

```
Subject:  Re:  I-D key handling
> Previous draft uses optional
ciphers.
> Should we keep algorithm agility?
We should simplify interoperability
and avoid platform-specific
primitives.  Therefore, adopt fixed
AES key length.
```

Non-verbose email (quotes removed)

```
Subject:  Re:  I-D key handling
We should simplify interoperability
and avoid platform-specific
primitives.  Therefore, adopt fixed
AES key length.
```

- Non-verbose preprocessing removes lines starting with > to isolate the sender's own contribution.

Rationale

To simplify implementation across platforms, we avoid using platform-specific cryptographic primitives.



Decision

The protocol **SHOULD** use standardized AES encryption with a fixed key length.

Each encrypted payload includes an integrated authentication tag to confirm message integrity before decryption.

All decryption and verification failures **MUST** return a uniform error to reduce side-channel leakage.

Sensitive metadata is encrypted in the same envelope as user data to prevent leakage.

II. Baseline Retrievers

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Sparse Baseline

- **BM25**

Dense Baselines

- **Semantic**: retrieves candidates via nearest-neighbor similarity in embedding space.
- **ReasonIR**: a dense retriever designed to capture reasoning-intensive relevance.

General vs. Domain-Specific Baselines

- **General-purpose**: Promptriever (denoted as **P**) from MTEB.
- **Domain-specific**: SIGIR, trained on similar GitHub rationale–decision data. We denote it as **Mistral^R** (R = real, contrary to S = synthetic).

II. Retriever Fine-Tuning Setup

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- **Backbone model:** Mistral-7B.
- **Training data:** synthetic rationale–decision pairs from two LLMs (Qwen, Llama) and two email settings (non-verbose, verbose), yielding four variants: QN, QV, LN, and LV. We denote the fine-tuned model as **Mistral^S**, to differentiate it from **Mistral^R** (R = real; S = synthetic).
- **Email evaluation sets:** RFC 7657, RFC 8205, RFC 8335, and a combined set (All RFCs).
- **External benchmark:** A similar dataset sourced from prior work on GitHub: development split for model selection; test and OOD splits for cross-distribution comparison.

Split	BM25	P	M^R	M^S (mean $\pm$ sd)
Test	63.23	67.76	79.27	71.93 $\pm$ 1.51
OOD	41.49	46.71	57.16	48.24 $\pm$ 1.47
Macro-avg	52.36	57.24	68.22	60.09

Table: MRR@10 on GitHub. M: Mistral; M^R : trained on real data; M^S : trained on synthetic data.

On GitHub, **Mistral^R** is the strongest retriever across both test and OOD splits.

Macro MRR@10

- M^R : 68.22
- M^S : 60.09
- Promptriever: 57.24
- BM25: 52.36

Relative gain of M^R

- vs BM25: +15.86
- vs Promptriever: +10.98
- vs M^S : +8.13

Interpretation: real-data supervision is better aligned with GitHub-style rationale–decision retrieval.

Email	BM25	P	M^R	M^S (mean $\pm$ sd)
RFC 7657	54.33	60.60	60.30	61.25 $\pm$ 2.26
RFC 8205	73.91	70.83	81.96	80.73 $\pm$ 1.84
RFC 8335	56.74	46.41	46.52	61.13 $\pm$ 6.48
All RFCs	60.35	59.23	62.22	65.60 $\pm$ 1.75
Macro-avg	61.33	59.27	62.75	67.18

Table: MRR@10 on email data. M: Mistral; M^R : trained on real data; M^S : trained on synthetic data.

On email, **Mistral^S** generally performs best and achieves the highest macro score.

II. Results: Email (Takeaway)

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Macro MRR@10

- M^S : **67.18**
- M^R : 62.75
- BM25: 61.33
- Promptriever: 59.27

Per-RFC winner

- RFC 7657: M^S
- RFC 8205: M^R
- RFC 8335: M^S (largest gap)
- All RFCs: M^S

Interpretation: synthetic training better captures email-specific rationale patterns.

Next, we highlight the key differences between **Mistral^R** and **Mistral^S**.

III. Results: Variant Analysis

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- Empirically, training on Qwen-generated data, particularly the non-verbose (QN) variant, **Mistral**^S (QN), yields the best retrieval scores overall, whereas manual inspection suggests that Llama's outputs often read closer to RFC style.
- This contrast indicates that *stylistic fidelity to RFC prose is not, by itself, the key driver of retrieval effectiveness*. Instead, properties such as lexical clarity, concise phrasing, and well-structured presentation of salient anchors (identifiers, error terms, actionable statements) appear to align better with our retriever.

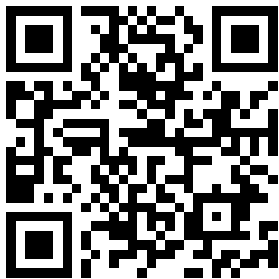
Based on the observed failure patterns, we infer that **Mistral**^R prioritizes broad semantic similarity over explanatory alignment, which leads to over-reliance on surface phrasing and missed rationale signals. In contrast, **Mistral**^S (QN) appears to better adapt to the variety of rationale types present in emails, likely due to the greater complexity and larger size of the email data.

In summary, our contributions are as follows:

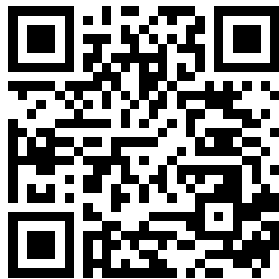
- Design Decision Recovery: a specialized prompting strategy that enables LLMs to infer and reconstruct missing design decisions from technical email discussions.
- RFCAlign: a semi-synthetic rationale-to-decision dataset comprising around 350 000 samples.
- RFC-DRAAlign: Using RFCAlign, we fine-tune Mistral-7B for rationale–decision (R–D) retrieval, establishing a new benchmark for this task.

- **Limited evaluation scale:** test sets remain relatively small.
- **Constrained retrieval setting:** current experiments assume retrieval from a fixed candidate pool.
- **Future direction:** extend to open-archive retrieval; reformulate design decisions in an explanation-oriented style to guide the retriever with explicit rationale-type signals (e.g., system design, performance optimization, risk mitigation, and security considerations).

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GitHub



Dataset